JEE NEXUS • NCERT CLASS 12 • PHYSICS

Chapter 3: Current Electricity

NCERT: Current Electricity | Avg PYQs: 2 to 3 Questions per Shift | Weightage: 8–12 Marks | Diff: Moderate

& HIGH REPEAT CHAPTER (MUST DO)

HIGH-YIELD JEE SUBTOPICS:

Kirchhoff's Laws & Wheatstone Bridge • Drift Velocity & Mobility Relations • Cell Combinations & EMF vs Terminal Potential • Galvanometer to Ammeter/Voltmeter Conversion • Meter Bridge & Temperature Dependence of Resistivity

NTA Focus: Direct numerical value questions on circuit loop currents and galvanometer coil resistance calculations.

1. Microscopic Ohm's Law & Drift Mechanics

In conductors, free electrons execute random thermal motion with velocities around 10^5 m/s, but net thermal current is zero due to isotropic velocity distribution.

Upon applying an external electric field E, electrons experience acceleration -eE/m and acquire an average steady drift velocity v_d opposing the electric field between successive relaxation collisions.

MOST REPEATED PYQ TOPIC (80% JEE Questions)

Ø=Ý% NTA Favorite: Drift Velocity & Current Density Relations

Always remember the link between microscopic and macroscopic quantities. Current I through a conductor of cross-section A is given by:

$$I = n e A v_d \quad \implies \quad J = \frac{I}{A} = n e v_d = \sigma E = \frac{E}{\rho}$$

- Drift velocity magnitude: v_d = (e E Ä) / m
- Mobility ¼ = v_d / E = (e Ä) / m (Always positive, SI unit: m²/(V·s))
- Conductivity Ã = n e² Ä / m and Resistivity Á = m / (n e² Ä)

COMMON JEE TRAP / SILLY MISTAKE ALERT

&þ Common Student Trap: Stretching a Wire vs Cutting

When a wire of initial resistance R is STRETCHED by n times its length (or radius halved), VOL
L alone!

$$R = \rho \frac{L}{A} = \rho \frac{L^2}{\text{Volume}} \implies R_{\text{new}} = n^2 R_{\text{initial}}$$

- If stretched so length increases by x% (where x < 5%): "R/R "H 2("L/L)
- If stretched so radius decreases by 1%: "R/R "H -4("r/r) = +4% increase

2. Kirchhoff's Laws & Wheatstone Bridge Archetypes

Kirchhoff's Current Law (KCL) is based on the Conservation of Charge (£ I_in = £ I_out). Ki
Conservation of Energy (£ "V = 0 around any closed mesh).

A Wheatstone bridge consists of four resistors. When the galvanometer shows zero deflection (null point), the bridge is balanced and no current flows through the central branch.

FORMULA & SHORTCUT VAULT

Ø=ÜÐ Master Formula Vault: Wheatstone Condition & Maximum Sensitivity

For null deflection in bridge ABCD with resistances P, Q, R, S:

$$\frac{P}{Q} = \frac{R}{S} \quad \iff \quad V_B = V_D$$

- Galvanometer and battery can be interchanged without affecting balance condition.
- The bridge is most sensitive when all four resistances P, Q, R, S are of approximately equal magnitude.

DIRECT NCERT STATEMENT / EXCEPTION

Ø=ÜØ Direct NCERT Statement: Carbon Resistor Color Code & Tolerances

NCERT Table 3.2: "BBROY Great Britain Very Good Wife" — Black (0), Brown (1), Red (2), Orange (3), Yellow (4), Green (5), Blue (6), Violet (7), Grey (8), White (9). Gold (±5%), Silver (±10%), No Color (±20%).

METHOD / DERIVATION: Step-by-Step Nodal Analysis for Complex Multi-Loop Circuits

[Step 1] Identify all essential nodes and assign one node as Reference/Ground (V = 0 V).

[Step 2] Assign unknown node potentials V1, V2 to the remaining nodes.

[Step 3] Write KCL at each node: sum of outward currents = 0: £ (V_node - V_neighbor) / R = 0.

[Step 4] Solve linear algebraic equations directly using substitution or Cramer's Rule.

3. Cells, Combinations & Meter Bridge

An electrochemical cell maintains a potential difference between its electrodes. The EMF E is the potential difference between terminals in an open circuit (zero current).

When drawing current I, terminal potential V = E - I·r due to internal resistance r. During charging by an external charger, V = E + I·r.

FORMULA & SHORTCUT VAULT

Ø=ÜÐ Master Formula Vault: Mixed Cell Combinations & Max Power

For n identical cells in m parallel rows with external load R:

$$I = \frac{n E}{R + \frac{n r}{m}}, \quad P_{\text{max}} = \frac{n m E^2}{4 r} \quad \text{when } R = \frac{n r}{$$

- Parallel cells of unequal EMF: E_eq = (E1/r1 + E2/r2) / (1/r1 + 1/r2)
- Equivalent internal resistance: 1/r_eq = 1/r1 + 1/r2

MOST REPEATED PYQ TOPIC (80% JEE Questions)

Ø=Ý% NTA Favorite: Galvanometer Conversion to Ammeter & Voltmeter

1. Ammeter: Connect small shunt resistance S in PARALLEL with galvanometer G: S = (I_g · G) / (I - I_g)
2. Voltmeter: Connect high resistance R in SERIES with galvanometer G: R = (V / I_g) - G

$$S = \frac{I_g G}{I - I_g} \quad \text{and} \quad R = \frac{V}{I_g} - G$$

- Ideal Ammeter resistance = 0 ©. Ideal Voltmeter resistance = " ©.
- If range is increased n times (I = n I_g): S = G / (n - 1)
- If voltmeter range is increased n times (V = n V_g): R = (n - 1) G

Top Solved PYQ Archetypes (2)

Archetype #1 (JEE Main 2024 / 2023 Repeating Archetype)

Concept: Galvanometer to Multi-Range Ammeter & Shunt Calculation

A galvanometer with coil resistance 50 © gives full-scale deflection for a current of 10 mA. T reading up to 5 A, find the value of shunt resistance required.

Methodical Solution Steps:

- Given: Galvanometer resistance G = 50 ©, Full-scale deflection current I_g = 10 mA = 0.01 A.
- Total current to be measured I = 5 A.
- Current passing through shunt S: I_s = I - I_g = 5 - 0.01 = 4.99 A.
- Since shunt and galvanometer are in parallel, V_s = V_g: S · (I - I_g) = I_g · G.
- S = (0.01 × 50) / 4.99 = 0.50 / 4.99 "H 0.1002 ©.

NTA Trap & Shortcut:
Shortcut: S = G / (n - 1) where n = I / I_g = 5000 / 10 = 500. S = 50 / 499 "H 0.1002 ©. Beware of units: 1

Archetype #2 (JEE Main 2023 / 2021 Numerical)

Concept: Unequal Cells in Parallel with Internal Resistance

Two cells of EMF 2 V and 4 V with internal resistances 1 © and 2 © respectively are connecte Calculate the equivalent EMF and terminal potential across an external load of 5 ©.

Methodical Solution Steps:

- Formula for two parallel cells: E_eq = (E1/r1 + E2/r2) / (1/r1 + 1/r2)
- E_eq = (2/1 + 4/2) / (1/1 + 1/2) = (2 + 2) / (1.5) = 4 / 1.5 = 8/3 V "H 2.67 V.
- Internal resistance r_eq: 1/r_eq = 1/1 + 1/2 = 3/2 !Ò r_eq = 2/3 ©.
- Circuit current through 5 © load: I = E_eq / (R + r_eq) = (8/3) / (5 + 2/3) = (8/3) / (17/3) = 8/17 A.
- Terminal potential V = I × R = (8/17) × 5 = 40/17 V "H 2.35 V.

NTA Trap & Shortcut:
Common mistake: Students directly average EMFs (2 + 4)/2 = 3 V, which ignores internal resistance weighting!

Final Answer: E_eq = 2.67 V (8/3 V); V_t = 2.35 V (40/17 V)

NCERT Hidden Gems & Tables (2)

Manganin and Constantan for Standard Resistors (NCERT Class 12 Part 1, Page 104)

"NCERT Section 3.7: Alloys such as Manganin, Constantan, and Nichrome are used for standard wire-wound resistors because their resistivity is very high and has extremely low temperature coefficient of resistance (almost independent of temperature)."

Why NTA Asks This: Asked directly as a statement/assertion-reason question in JEE Main 2022 & 2024.

Semiconductor vs Metallic Resistivity Curve (NCERT Class 12 Part 1, Page 105 (Fig 3.9 & 3.10))

"For metals, resistivity increases with temperature (positive ±). For semiconductors and insulato with increasing temperature because charge carrier density n increases exponentially: n(T) " exp

Why NTA Asks This: NTA frequently asks students to match the correct Á vs T graph for Copper, Nichrom

Checklist & Personal Notes

Must-Solve Subtopic Checklist:

- [ ] Derive Drift Velocity v_d = eEÄ/m and Current Relation I = n e A v_d
- [ ] Solve Balanced & Unbalanced Wheatstone Bridge with Symmetrical Fold Method
- [ ] Calculate Terminal Voltage during Charging vs Discharging: V = E + Ir vs V = E - Ir
- [ ] Convert Galvanometer (G) to Ammeter of range I and Voltmeter of range V
- [ ] Calculate Maximum Power Transfer across variable load (R = r_eq)
- [ ] Analyze temperature coefficient of resistance R(T) = R_0[1 + ±(T - T_0)]

My Chapter Scratchpad / Student Notes:

Remember shunt is parallel, multiplier resistor is in series. V = E - Ir.